

IFT-Guided Profile Likelihood Scan

Algorithm Description

Newtrinos.jl

Motivation

Profile likelihood scans require solving a separate optimisation at each grid point ϑ_k :

$$\hat{\nu}(\vartheta_k) = \arg \max_{\nu} [\log \mathcal{L}(\vartheta_k, \nu) + \log \pi(\nu)], \quad (1)$$

where $\mathcal{L}$ is the likelihood and $\pi(\nu)$ the nuisance prior. For experiments with $\mathcal{O}(40\text{--}90)$ nuisance parameters, each optimisation is expensive and a poor starting point can trap the optimiser in a local minimum.

The IFT-guided scan uses a local Gaussian (MGVI) approximation of the posterior at the current optimum to *predict* the nuisance optimum at the next grid point, avoiding cold restarts.

1 Prior-Normalised Coordinates (z -space)

Every nuisance parameter ν_i with prior π_i is mapped to

$$z_i = \Phi^{-1}(F_{\pi_i}(\nu_i)), \quad (2)$$

where F_{π_i} is the prior CDF and Φ^{-1} the standard-Normal quantile function. Collecting all nuisance parameters gives $\mathbf{z} \in \mathbb{R}^n$ with prior $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

The scan parameter ϑ_k is fixed to a point mass at each grid value and is *excluded* from $\mathbf{z}$; it never enters the z -space optimisation.

At grid point ϑ_k the z -space posterior is

$$\log p_k(\mathbf{z}) = \log \mathcal{L}(\vartheta_k, f(\mathbf{z})) - \frac{1}{2} \|\mathbf{z}\|^2 + \text{const}, \quad (3)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ maps back to physical nuisance space.

2 Local MGVI Approximation

After optimising at grid point ϑ_{k-1} we obtain $\mathbf{z}_{k-1}^*$. Assuming the likelihood arises from a forward model $\mu(\vartheta, \nu)$ with observation-noise Fisher information $\mathbf{F} \in \mathbb{R}^{m \times m}$, the MGVI posterior precision at the current optimum is

$$\mathbf{\Lambda} = \mathbf{J}^\top \mathbf{F} \mathbf{J} + \mathbf{I}_n, \quad \mathbf{J} = \frac{\partial \mu}{\partial \mathbf{z}} \in \mathbb{R}^{m \times n}, \quad (4)$$

and the local posterior covariance is $\mathbf{\Sigma} = \mathbf{\Lambda}^{-1}$. The Gaussian approximation centred at the current optimum is

$$q_{k-1}(\mathbf{z}) \approx \mathcal{N}(\mathbf{z}_{k-1}^*, \mathbf{\Sigma}). \quad (5)$$

3 IFT Predictor Step

To predict the optimum at the *next* scan point ϑ_k , evaluate the gradient of the new posterior at the old optimum:

$$\mathbf{g}_k = \nabla_{\mathbf{z}} \log p_k(\mathbf{z}) \Big|_{\mathbf{z}=\mathbf{z}_{k-1}^*}. \quad (6)$$

This gradient is nonzero because shifting ϑ changes the likelihood, so the old optimum is no longer stationary. A single Newton step with the MGVI Hessian approximation ($\mathbf{\Lambda} \approx \mathbf{\Sigma}^{-1}$) gives the predicted new optimum:

$$\boxed{\mathbf{z}_k^{\text{pred}} = \mathbf{z}_{k-1}^* + \mathbf{\Sigma} \mathbf{g}_k.} \quad (7)$$

Why this works. For a quadratic log-posterior (Gaussian likelihood, linear forward model), the prediction is *exact*: one Newton step reaches the true new optimum regardless of the step size $\vartheta_k - \vartheta_{k-1}$, as verified by the toy Gaussian validation script. For non-quadratic likelihoods the prediction is approximate but provides a far better starting point than the neighbour’s $\mathbf{z}_{k-1}^*$, especially when the nuisance optimum varies smoothly across the scan.

4 Polish Step

The predicted $\mathbf{z}_k^{\text{pred}}$ initialises a gradient-based local optimisation (L-BFGS) in z -space:

$$\mathbf{z}_k^* = \arg \max_{\mathbf{z}} \log p_k(\mathbf{z}), \quad \text{initialised at } \mathbf{z}_k^{\text{pred}}. \quad (8)$$

Starting near the true optimum, the optimiser requires far fewer gradient evaluations than a cold start, reducing total cost and the probability of converging to a spurious local maximum.

5 BFS Traversal Order

Grid points are visited in *breadth-first order* starting from the grid index nearest to a user-supplied seed (typically the global best-fit point). This guarantees every point is warm-started from a direct neighbour whose posterior is maximally similar, rather than from a point several steps away. For a 2-D grid (e.g. ϑ_1 vs. ϑ_2), the BFS expands outward ring by ring from the seed, visiting each point once.

6 Algorithm Summary

Algorithm: IFT-guided profile likelihood scan

1. Build prior-normalising transform f and its inverse f^{-1} .
2. Sort grid points $\{\vartheta_k\}$ in BFS order from the seed index.
3. For each scan point ϑ_k in BFS order:
 - (a) Fix $\vartheta = \vartheta_k$; define $\log p_k(\mathbf{z}) = \log \mathcal{L}(\vartheta_k, f(\mathbf{z})) - \frac{1}{2}\|\mathbf{z}\|^2$.
 - (b) If k is the BFS root: set $\mathbf{z}_{\text{init}} = f^{-1}(\boldsymbol{\theta}_{\text{seed}})$.
 - (c) Else:
 - (i) Retrieve parent optimum $\mathbf{z}_{\text{parent}}^*$.

- (ii) Compute MGVI covariance $\Sigma = (\mathbf{J}^\top \mathbf{F} \mathbf{J} + \mathbf{I})^{-1}$ at $\mathbf{z}_{\text{parent}}^*$ under p_{parent} .
 - (iii) Evaluate IFT gradient $\mathbf{g}_k = \nabla_{\mathbf{z}} \log p_k(\mathbf{z}_{\text{parent}}^*)$.
 - (iv) Set $\mathbf{z}_{\text{init}} = \mathbf{z}_{\text{parent}}^* + \Sigma \mathbf{g}_k$.
 - (d) Polish: $\mathbf{z}_k^* = \text{L-BFGS}(\log p_k, \mathbf{z}_{\text{init}})$.
 - (e) Store $\hat{\nu}(\vartheta_k) = f(\mathbf{z}_k^*)$ and $\hat{\ell}(\vartheta_k) = \log \mathcal{L}(\vartheta_k, \hat{\nu}(\vartheta_k))$.
-

7 Computational Cost

Let n be the number of nuisance parameters and m the data dimension.

- **MGVI covariance** ($\mathcal{O}(mn^2 + n^3)$ per grid point): one forward-model Jacobian and one matrix inversion.
- **IFT gradient**: one additional likelihood gradient evaluation ($\mathcal{O}(mn)$ with reverse-mode AD or $\mathcal{O}(mn/B)$ batches with ForwardDiff chunk size B).
- **Polish**: typically 5–30 $\times$ fewer L-BFGS iterations than a cold start, because $\mathbf{z}_k^{\text{pred}}$ is already near the optimum.

For $n \gtrsim 20$ (where ForwardDiff requires multiple chunked passes) the reduction in polish iterations typically outweighs the MGVI overhead, giving a net speed-up of 2–5 $\times$ over sequential warm-starting alone.